

Name again: My problems are compounding continuously! _____

Period _____

This is the calculator page! Turn in your other sheet before taking out your calculator

$$FV = C \left[\frac{(1+i)^n - 1}{i} \right] \quad PV = C \left[\frac{1 - (1+i)^{-n}}{i} \right]$$

7. You have two different investment opportunities that you are trying to choose between.

Option 1: Invest \$3000 in a CD (certificate of deposit) that grows at a 4% annual interest rate (compounded monthly). The CD requires you to leave it “untouched” for 5 years.

Option 2: Invest \$60 every month for 5 years at an annual interest rate of 3%.

- a) Both options involve investing a total of \$3000 over 5 years. Find the Future Value for both options. [4 pts.]

FV Option 1 _____

FV Option 1 _____

- b) How would the annual interest rate have to change in “Option 1” so that the two investments would have the same Future Value? Round your interest rate to the nearest hundredth place. [3 pts]
- c) Give a financial reason that you might actually choose the investment with the SMALLER Future Value. [0 points. Obviously don’t spend too much time thinking about this one!]

Turn over for the last problem!

8. You have your eye on a \$35,000 used Dodge Charger car, and will pay for it by paying partly with cash (putting money down) and partly with financing (taking a loan). The car dealership is offering 5% annual interest rate on a 3-year loan. You know that you can only afford \$500 monthly payments. How much cash would you have to put down to stay within your monthly budget? [5 pts].